WORKED SOLUTION

STEP I 2017, Question 5

Pure • Optimisation by Differentiation

STEP I 2017, Question 5

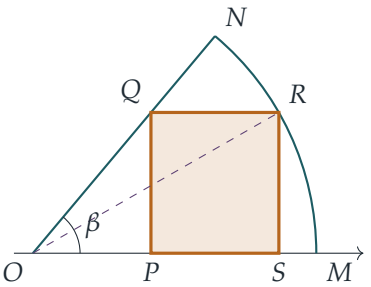
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q5

The key idea

Once you place the vertices, everything is fixed by the single variable x: Q sits directly above P on the line $y = x\tan\beta$, and R is level with Q on the circle. Pythagoras gives s in terms of x, and the area becomes an ordinary one-variable function to differentiate. The final angle condition then drops out of the half-angle formula for tan.

Solution



The area

Since $PQRS$ is a rectangle with P and S on the x-axis, the sides PQ and SR are vertical. So Q has x-coordinate x. It lies on ON, the line $y = x\tan\beta$, so $Q = (x,\, x\tan\beta)$, and $R = (s,\, x\tan\beta)$. The rectangle has width $PS = s - x$ (positive, as S is further from O than P) and height $PQ = x\tan\beta$, so

$$A = x(s - x)\tan\beta,$$

as required.

Finding s and $\frac{\mathrm{d}A}{\mathrm{d}x}$

Since R lies on the circle, $OR = a$, so $s^2 + x^2\tan^2\beta = a^2$. As $s > 0$,

$$s = \sqrt{a^2 - x^2\tan^2\beta}.$$

Differentiating $s^2 = a^2 - x^2\tan^2\beta$ implicitly gives $2s\dfrac{\mathrm{d}s}{\mathrm{d}x} = -2x\tan^2\beta$, so $\dfrac{\mathrm{d}s}{\mathrm{d}x} = -\dfrac{x\tan^2\beta}{s}$. Now write $A = (xs - x^2)\tan\beta$ and use the product rule on xs:

$$\begin{aligned}\frac{\mathrm{d}A}{\mathrm{d}x} &= \left(s + x\frac{\mathrm{d}s}{\mathrm{d}x} - 2x\right)\tan\beta \\ &= (s - 2x)\tan\beta - \frac{x^2}{s}\tan^3\beta,\end{aligned}$$

which is the required result.

The greatest area

First the range of x. As x increases from 0, Q moves up ON and R moves round the arc towards N; the rectangle shrinks to a vertical segment when R reaches N, that is when $x = a\cos\beta$ (then $s = \sqrt{a^2 - a^2\sin^2\beta} = a\cos\beta = x$). So $0 < x < a\cos\beta$, and A is a continuous function on $[0, a\cos\beta]$ with $A = 0$ at both ends and $A > 0$ in between. Its greatest value is therefore at a stationary point inside the interval.

Setting $\frac{\mathrm{d}A}{\mathrm{d}x} = 0$, dividing by $\tan\beta > 0$ and multiplying by $s > 0$:

$$\begin{aligned}s^2 - 2xs &= x^2\tan^2\beta \\ s^2 - 2xs + x^2 &= x^2(1 + \tan^2\beta) = x^2\sec^2\beta \\ (s - x)^2 &= x^2\sec^2\beta.\end{aligned}$$

Ask yourself Which square root should we take, and how do we know this is a maximum?

Both $s - x$ and $x\sec\beta$ are positive (since $0 < \beta < \frac{\pi}{2}$), so $s - x = x\sec\beta$; the other root would make $s - x$ negative. So there is exactly one stationary point in $0 < x < a\cos\beta$, given by

$$s = x(1 + \sec\beta),$$

and by the previous paragraph it must give the greatest area.

To find that area, substitute into $s^2 + x^2\tan^2\beta = a^2$. Using $\tan^2\beta = \sec^2\beta - 1$,

$$(1 + \sec\beta)^2 + \tan^2\beta = 2\sec\beta + 2\sec^2\beta = 2\sec\beta\,(1 + \sec\beta),$$

so

$$x^2 = \frac{a^2}{2\sec\beta\,(1 + \sec\beta)} = \frac{a^2\cos^2\beta}{2(1 + \cos\beta)}.$$

At this point $s - x = x\sec\beta$, so

$$A = x \cdot x\sec\beta \cdot \tan\beta = x^2\,\frac{\sin\beta}{\cos^2\beta} = \frac{a^2\sin\beta}{2(1+\cos\beta)}$$
$$= \frac{a^2 \cdot 2\sin\tfrac{1}{2}\beta\cos\tfrac{1}{2}\beta}{2\cdot 2\cos^2\tfrac{1}{2}\beta} = \tfrac{1}{2}a^2\tan\tfrac{1}{2}\beta,$$

using $\sin\beta = 2\sin\frac{1}{2}\beta\cos\frac{1}{2}\beta$ and $1+\cos\beta = 2\cos^2\frac{1}{2}\beta$.

The angle ROS

In the right-angled triangle ORS, $\tan\angle ROS = \dfrac{SR}{OS} = \dfrac{x\tan\beta}{s}$. At the maximum,

$$\tan\angle ROS = \frac{\tan\beta}{1+\sec\beta} = \frac{\sin\beta}{\cos\beta+1} = \tan\tfrac{1}{2}\beta,$$

by the same two half-angle identities. Both $\angle ROS$ and $\frac{1}{2}\beta$ lie in $(0, \frac{\pi}{2})$, where tan is one-to-one, so $\angle ROS = \frac{1}{2}\beta$: at the best rectangle, OR bisects the sector.

Checks

- Take $a = 1$, $\beta = \frac{\pi}{3}$. Then $x^2 = \frac{1/4}{3} = \frac{1}{12}$, $s = 3x$, and $A = x\cdot 2x\cdot\sqrt{3} = \frac{\sqrt{3}}{6} \approx 0.2887$, which equals $\frac{1}{2}\tan\frac{\pi}{6}$. A computer search over a fine grid of x, for many random a and β, found the same maximum and the same optimal x.
- As $\beta \to 0$ the sector collapses and $\frac{1}{2}a^2\tan\frac{1}{2}\beta \to 0$, as it should.
- Units: A is a^2 times a pure number, so it is an area.

Common slips

- Forgetting that s depends on x, and differentiating $x(s-x)\tan\beta$ as if s were constant.
- Taking $s - x = -x\sec\beta$ as well, or not justifying that the stationary point is a maximum.

Beyond STEP

The tidy answer, with OR along the bisector of the sector, has a neat explanation: if $R = (a\cos\theta, a\sin\theta)$, a short calculation gives $A = \frac{1}{2}a^2\big(\cos(2\theta-\beta) - \cos\beta\big)/\sin\beta$, which is plainly greatest when $\theta = \frac{1}{2}\beta$. Choosing the right variable can make an optimisation almost trivial. At university you meet constrained optimisation in several variables, where instead of eliminating s you could maximise $x(s-x)$ subject to $s^2 + x^2\tan^2\beta = a^2$ using Lagrange multipliers. Siklos, *Advanced Problems in Mathematics*, has many more questions in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.